

AI Exposure at Three Levels: Reconciling Occupation, Task and Skill Measures of the Labour Market

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Contents

Extended summary Introduction Section 1: Three descriptions of one labour market Section 2: Measures collected at three levels Section 3: Reconciling the measures Section 4: What the instrument will bear Section 5: Realised wages and employment Conclusion References

Extended summary

Exposure to generative artificial intelligence is measured at three different levels of the labour market. Felten, Raj and Seamans (2021) score the fifty-two abilities of the Occupational Information Network. Eloundou, Manning, Mishkin and Rock (2023) score its task statements. Srinivasan, Chen and Zakerinia (2024) score occupations. Applied work uses each of them at whatever level it requires, and the aggregation or disaggregation that carries a score to that level has not been validated.

This paper puts the measures on a common footing and reports four results.

The propagation from tasks to skills can be checked against an independent measurement at the skill level. Carrying the Eloundou task ratings through the Network’s activity crosswalk reproduces Felten’s ability-level scores at a Pearson correlation of 0.667 for reading comprehension and 0.546 for language modelling, against 0.289 for real-time video games and 0.172 for image recognition. Eloundou’s own human and machine raters agree at 0.591, which places the propagation at the reliability of its source and shows it discriminating between AI applications in the right direction.

The capability measures are less independent than their agreement implies. The Srinivasan automation score is the Eloundou construction applied to a later release of the Network with a later version of the same model, using the same rubric and weights. Their correlation of 0.833 across occupations establishes that the procedure reproduces across vintages, which is a weaker claim than convergence between approaches.

Augmentation, as currently measured, is mostly a transformation of automation. Srinivasan and co-authors define it as one minus a Herfindahl index over the shares of exposed and unexposed tasks, which is a quadratic in the exposed share, and a quadratic in the automation score accounts for 77 percent of its variance. Net of that, skill groups explain 0.209 of the residual and eleven of twenty-five groups load on it, with social skills, engineering and technology, arts and humanities, work output and psychomotor abilities on the augmenting side. Business and management and mental processes do not survive, falling from statistics near three to 0.4 and 1.2.

Realised wages cannot settle whether any of this has occurred. Across a balanced panel of 693 occupations from 2012 to 2025, no skill group moved by more than 1.51 times its own year-to-year volatility after 2022, and the observed period sits inside the range produced by relabelling ordinary years. Every deviation from the pre-2020 path begins in 2020 or 2021, before large language models were released.

The Network constrains all of this in four documented respects. Its importance ratings compress towards uniformity when converted into shares. Its descriptors are collinear enough that the effective dimension of the space is about a third of its nominal dimension. Its two descriptions of what an occupation involves agree at a median rank correlation of 0.33. And its occupational content is revised too slowly to register a job changing while keeping its name. It is used here because it is free, covers the whole economy, and supplies the vocabulary in which every measure above is expressed.

Introduction

The empirical literature on generative artificial intelligence and work rests on a small number of exposure measures. Those measures have been compared with each other, and their disagreements noted, most often as a question of whether capability ratings or observed usage is the more informative. Less attention has been paid to the fact that they were collected at different levels of description, so that comparing them requires a translation whose properties are unexamined.

The translation matters because the levels differ in what they can support. Wages are observed once per occupation, and there are 829 of them in the 2025 Occupational Employment and Wage Statistics, carrying a wage bill of 10.70 trillion dollars. There are 18,796 task statements, so a set of task values has more unknowns than facts and cannot be contradicted by the data producing it. There are 161 descriptors, and after grouping twenty-five, so a set of skill values can fail. Whether a quantity in this literature is testable depends on how many task statements the Network's analysts wrote. Freund and Mann (2026) cluster roughly nineteen thousand tasks into thirty-eight categories before estimating, which is the step that renders their estimation feasible.

Section 1 sets out the three layers. Section 2 places each measure at the level where it was collected. Section 3 reconciles them. Section 4 reports what the underlying data will support. Section 5 applies the apparatus to fourteen years of wages and employment.

Section 1: Three descriptions of one labour market

An industry produces output, that output is produced by activities, and those activities draw on human capability, so occupations, tasks and skills describe one process at three distances. The premise runs from Autor, Levy and Murnane (2003) and Acemoglu and Autor (2011) through the skill literature descending from Katz and Murphy (1992). Freund and Mann (2026) give it a general equilibrium form in which occupations bundle tasks, workers hold task-specific skills, and automating a task raises the weight on the tasks that remain. Estimating that model on

individual worker data, they project that the return to social and non-routine manual skills rises and the return to analytical skills falls.

The three layers do different work. The wage bill is a budget: quantities constructed as shares of a total return that total whatever the shares are, so a reconciliation to the national wage bill tests nothing. Tasks are where the technology has been observed, since every measure of generative AI was built by judging a capability against a description of work. Skills are where a claim can fail, being fewer in number than the occupations whose wages are observed.

The skill grouping used throughout is the third level of the Network’s own content model, which gives twenty-five named groups across abilities, skills, knowledge and work activities. Figure 1 shows the three layers and the measures entering at each.

Section 2: Measures collected at three levels

Felten, Raj and Seamans (2021) built their measure by surveying mTurk respondents on the relatedness of ten AI applications to the fifty-two abilities of the Network, then combining the resulting matrix with occupational ability requirements. They publish the intermediate ability-level scores. Those abilities are fifty-two of the 161 descriptors used here, so the measure enters the skill layer without propagation. All fifty-two join after one renaming between releases. The ten applications are drawn from 2016-era progress benchmarks and include image recognition, real-time video games and instrumental track recognition, so the headline index measures exposure to artificial intelligence broadly. The application-by-ability matrix allows any single application to be isolated, and the language modelling and reading comprehension columns are the relevant ones for the present technology.

Eloundou, Manning, Mishkin and Rock (2023) rate 19,265 task statements on a four-point rubric distinguishing tasks a language model can substantially accelerate from those it cannot, with separate ratings by human annotators and by GPT-4. Of these, 18,179 join the task set used here. This is the measure Freund and Mann employ. Reaching skills requires the Network’s crosswalk from task statements to detailed work activities, which roll up to the forty-one generalized activities and thence, through the published crosswalks from skills and abilities to work activities, to the wider descriptor set.

Srinivasan, Chen and Zakerinia (2024) score automation and augmentation for 911 occupations, of which 910 join the Network by name, giving 785 six-digit occupations and covering 681 of the 693 in the panel used here. Both indices are built from task descriptions of Network release 25.1 classified by GPT-4o on the Eloundou rubric. Job postings enter their paper to validate the indices and not to construct them, so neither score contains labour market outcomes.

The Anthropic Economic Index classifies conversations with a deployed model against Network tasks and reports the composition of interaction modes for each. Its automation share is the sum of directive and feedback-loop modes and its augmentation share the sum of iteration, validation and learning, with a filtered residual completing one hundred percent, so the two shares are compositional and correlate minus 0.588 by construction. The index covers 20.9 percent of the

tasks considered here, and the covered tasks are more exposed by 0.236 on a zero to one scale than those it omits, with the most used tenth accounting for three quarters of recorded usage. It observes usage rather than capability and is used below as an outcome within its covered subset.

Section 3: Reconciling the measures

Two of the measures reach ability-level exposure by routes sharing no data. Felten and co-authors obtain it from an mTurk survey on applications and abilities. Propagating the Eloundou task ratings through the activity crosswalk obtains it from a language model’s reading of task text. Each of the ten Felten applications can be compared against the propagated series separately.

Propagated exposure correlates 0.667 with reading comprehension and 0.546 with language modelling, and 0.289 with real-time video games and 0.172 with image recognition. The ordering follows the closeness of each application to what a language model does. Eloundou’s human and GPT-4 raters agree on 65.6 percent of task labels and their numeric scores correlate 0.591, so the propagation performs at the reliability of its source on the applications that concern it and well below on those that do not.

Variation survives the aggregation. Averaging eighteen thousand task ratings into twenty-five groups gives a coefficient of variation of 0.292 across a range from 0.045 to 0.513, with mental processes highest, foundational literacy and numeracy close behind, and physical abilities lowest.

The Srinivasan automation score correlates 0.833 with Eloundou exposure aggregated to occupations. Their score applies the Eloundou rubric to Network release 25.1 using GPT-4o, weighting the two exposed categories one and one half and weighting each task by its importance. The correlation therefore measures reproducibility across data vintages and model versions. Several capability measures in this literature share a rubric or a source of task descriptions, and agreement among them should be read accordingly.

Augmentation requires the same care. Srinivasan and co-authors define it as one minus a Herfindahl index over the shares of exposed and unexposed tasks in an occupation, on the reasoning that a job combining automatable work with work that resists automation gains most. That index is a quadratic in the exposed share, and the automation score is a related summary of the same classification, so a quadratic in automation accounts for 77 percent of the variance in augmentation across occupations.

Inverting the two occupational scores onto the twenty-five skill groups requires no employment weighting, since exposure is not a share of the wage bill, so the effective sample remains at the number of occupations. Skill groups account for occupational exposure with cross-validated fit of 0.712 on automation and 0.415 on augmentation, and the named groups outperform ten principal components on both.

On the raw scores, ten of the twenty-five groups show an augmentation loading distinguishable from their automation loading, stable across five hundred bootstrap resamples. Removing the quadratic in automation and inverting the residual leaves eleven groups distinguishable, with skill groups accounting for 0.209 of the residual variation, and the two profiles correlate 0.604.

Social skills, engineering and technology, arts and humanities, work output and psychomotor abilities hold up. Business and management falls from a statistic of minus 3.1 to plus 0.4 and mental processes from minus 2.9 to minus 1.2. Figure 2 shows both profiles. Work involving people, physical execution and applied technical judgement is associated with the complementary side of these measures beyond what the construction implies. The apparent concentration of automation on cognitive and administrative content is the Herfindahl index restating an exposure share.

Section 4: What the instrument will bear

Importance in the Network is recorded on a five-point scale of which respondents use a narrow part, and converting such a rating into a share yields something close to uniform. In the median occupation the most important task rates 1.41 times the least important. Where the descriptor mix of a task is set to its occupational average, all 124,614 occupation by descriptor pairs containing more than one task carry a single distinct value, and the skill layer becomes a fixed linear image of one number per occupation. A descriptor proximity matrix built from occupational similarity places 98.5 percent of its 798,342 pairs above 0.90, with a coefficient of variation of 0.022. An occupation's profile across the forty-one generalized work activities sits a total variation distance of 0.10 from a uniform profile, and the Network's two descriptions of what a job involves agree at a median rank correlation of 0.33.

Valuing 17,383 tasks by allocating each occupation's wage bill in proportion to task importance produces values correlating 0.997 on ranks with dividing that wage bill by the occupation's number of tasks, with eighty-seven of the largest hundred tasks identical under the two procedures. The Network publishes task frequency over seven bands, which varies about fifteen times more than importance with a median ratio of 14.78, and substituting it lowers the correlation to 0.883.

Fitting occupational wages with descriptor intensities and evaluating on held-out occupations gives 0.554 for a single ordinal measure of required preparation, 0.700 for twenty principal components, 0.648 for the twenty-five named groups and 0.544 for all 161 descriptors. Employment weighting, which a wage bill decomposition requires, gives an effective sample of 121 against 820 occupations in the sense of Kish (1965), the twenty largest occupations carrying a third of employment.

Skill quantities are employment multiplied by intensity and involve no estimation. Price levels are not recoverable: eleven of twenty-five groups take a negative price in some year, and constraining prices to be non-negative pins twenty of twenty-five at exactly zero. Price changes are recoverable, and skill content explains 52 percent of the variation in occupational wage changes between 2022 and 2025.

Comparing any two periods in these data returns between five and nine of the twenty-five groups apparently changing course when nothing has happened. Conventional standard errors are computed against the precision of a window's average rather than against the movement of the series between ordinary years. Composition series are smooth enough that a shift worth 0.17 percent of a level per year produces a statistic of 8.2. The procedure adopted here estimates the

quantity for every year transition and builds the reference distribution by relabelling which three transitions are treated. Four results in the course of this work reached conventional significance and did not survive it.

Section 5: Realised wages and employment

The panel covers 693 occupations from 2012 to 2025, with a wage bill rising from 5.15 to 8.95 trillion dollars, assembled from the Occupational Employment and Wage Statistics and mapped across the 2018 revision of the occupational classification using the Network’s crosswalk, which raises coverage of the 2018 wage bill from 0.813 to 0.936.

Across 2022 to 2025 no skill group moved by more than 1.51 times its own year-to-year volatility. Five of twenty-five exceed two of their own standard errors, against a median of four across 286 relabellings. Quantities give a probability of 0.455 weighted by headcount.

A directional test against the Freund and Mann projection, with group membership assigned from their vocabulary before any outcome was examined, is not supported on wages and runs in the opposite direction, the returns to both social and analytical content having risen against their pre-period paths with analytical rising further. The social against analytical contrast in employment moves as projected, at a probability of 0.021 against all relabellings and 0.273 against consecutive triples, on the one specification carrying no pre-existing trend.

Figure 3 shows deviations from the pre-2020 path by year. On wages the residual runs at minus 0.005 in 2020 and reaches minus 0.025 by 2025. On employment the series takes an excursion of minus 0.040 in 2020 and remains positive from 2021. Large language models were released in late 2022, after both breaks. The pandemic sits between the pre-period and the treatment period and moves these series by more than the effect sought, so a before-and-after comparison in United States occupational data does not separate the technology from the recovery.

The pandemic moved which occupations grew. A technology of this kind should move what happens inside an occupation, which a comparison across occupations does not observe. The propagation validated in Section 3 is what a within-occupation design requires, and the Anthropic index supplies task-level outcomes on the fifth of tasks it covers, where selection falls on tasks rather than on time. Separately, the separation of augmentation from automation in Section 3 implies a shock structure absent from current models, in which some tasks are assisted rather than automated.

Conclusion

Three literatures measure exposure to generative artificial intelligence at three levels of one labour market. This paper states them together on public data, checks the propagation from tasks to skills against an independent measurement, and finds the capability measures closer to a single construct than their agreement suggests, with augmentation largely a transformation of automation and a residual that still carries skill structure. It documents four respects in which the Network constrains what may be concluded, and finds that the before-and-after design does

not identify an effect on realised wages, the pandemic break preceding the technological one in every series examined.

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